Magnetic bead-based immunoassay allows rapid, inexpensive and quantitative detection of human SARS-CoV-2 antibodies

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Supporting Information

Antigen purification. Expression and purification of recombinant SARS-CoV-2 Nucleocapsid protein (N-protein). A codon optimized synthetic gene expressing the full-length SARS-CoV-2 N-protein (Uniprot QHD43423.2) was obtained and cloned into pET28a by General Biosystems as described previously [1]. This plasmid was named pLHSarsCoV2-N and transformed into *E. coli* BL21 (λDE3) enabling the expression of N-terminal His-tagged N-protein after induction with IPTG. The cells were growth in 100 ml LB medium at 120 rpm at 37ºC to OD600nm of 0.3. The incubator temperature was lowered to 16ºC, after 30 min, IPTG was added to a final concentration 0.3 mM and the culture was kept at 120 rpm at 16ºC over/night. Cells were collected by centrifugation at 3,000 xg for 5 min. The cell pellet was resuspended in 10 ml of buffer 1 (Tris-HCl pH 8 50 mM and KCl 100 mM). Cells were disrupted by sonication on an ice bath. The soluble fraction was recovered after centrifugation at 20,000 xg for 10 min and loaded onto a 1 ml Histrap chelating Ni^{2+} column (Cytiva), which had been previously equilibrated with buffer 1. The column was washed with 10 ml of buffer 1 and bound proteins were eluted with buffer 1 containing increasing concentrations of imidazole. The His-tagged N-protein eluted at 300 mM. The fractions were pooled and buffer exchanged using a 5 ml desalting column (Cytiva) with 50 mM Tris-HCl pH 8, 100 mM KCl and 10% (v/v) glycerol as buffer. The final protein preparation was homogeneous as judged by SDS-PAGE analysis (Fig. S1). The purified protein was stored in aliquots at 4 ºC for one week or at -20 ºC for up to six months. The protein identity was confirmed by peptide mass fingerprint MALDI-TOF analysis (Fig. S1).

Human samples. Samples used by the team working in Brazil were collected from at the Complexo Hospital de Clínicas (CHC/UFPR) in Curitiba, Hospital Erasto Gaertner in Curitiba and Secretária Municipal de Saúde in Guaratuba. Samples for serological analysis comprised both serum and plasma-EDTA. Samples for point of care analysis were blood collected by finger stick puncture. COVID-19 positive cases were confirmed by the detection of SARS-CoV-2 RNA by real-time RT-PCR from nasopharyngeal sample swabs. Among the 90 COVID-positive cases there were: 40 mild, 25 hospitalized and 25 critical (intensive care unit). The time point of sampling ranged from 3 to 38 days after onset of symptoms: 14 samples were collected between 3-7 days of onset of symptoms, 23 samples were collected between 8-15 days of onset of symptoms. The remaining samples were collected more than 15 days after onset of symptoms. The cohort of 282 negative controls consisted of 235 pre pandemic samples collected in 2018, the remaining negative samples were collected from healthy donors, who declared no COVID-19 symptoms since December 2019. The Institutional Ethics Review Board of CHC/UFPR (n# 30578620.7.0000.0008) and CEP/HEG (n# 31592620.4.1001.0098) approved this study. Informed consent was obtained from all participants in this study. All methods were performed in accordance with the relevant guidelines and regulations.

For the work done in Germany, a total of 260 sera was used for method validation. 50 samples from healthy donors were purchased from Central BioHub GmbH (Henningsdorf, Germany) collected prior December 2019. The remaining 210 samples used in this study were collected during the ongoing pandemic after ethical review (179/2020/BO2, University Hospital Tübingen). 16 of these were from healthy donors, who reported no common cold symptoms during the pandemic to serve as a reference. 196 COVID-19 positive samples were from convalescent donor's post quarantine and were self-reported PCR-positive for SARS-CoV-2.

ELISA assays. The purified SARS-CoV-2 N-protein was diluted to 2 ng/µl in 50 mM Tris-HCl pH 8, 100 mM KCl, 10% (v/v) glycerol and 0.1 ml were added to the wells of polystyrene 96 well plates (Olen) overnight at 4 ^oC as described previously [1]. Unbound protein was removed using two washes with 0.2 ml of 1x TBST buffer pH 7.6 (Tris 2.42 $g.l^{-1}$, NaCl 8 $g.l^{-1}$ and tween 80 0.1% v/v). The wells were blocked with 0.2 ml 3% (w/v) skimmed milk in 1x TBST overnight at 4 ^oC. The serum was diluted in 1x TBST, 1% (w/v) skimmed milk (the dilution factors indicated in each experiment) and 0.1 ml were loaded onto the ELISA wells following by incubation at room temperature for 60 min. The wells were washed three times with 0.2 ml of 1x TBST followed by incubation with 0.1 ml of anti-human IgG-HPR from goat (Thermo Fisher - cat number 62-8420) diluted 1:3,000 in 1x TBST for 60 min. Wells were washed three times with 0.2 ml of 1x TBST. The HPR substrate TMB (Thermo Fisher - cat number 00-2023) was added to the wells (0.1 ml). The reaction was stopped after 10 min incubation at room temperature by addition of 0.1 ml 1 M HCl. The plates were put on the top of a white light transilluminator device and photographed. Samples were run in duplicates and data reported as average. The optical density was measured at 450 nm using a TECAN M Nano plate reader (TECAN) monochromator at bandwidth 9 nm and 25 flashes. The in house-built ELISA performed with accuracies similar to Euroimmun ELISA SARS-Cov-2 Nucleocapsid IgG commercially available. Details will be published elsewhere (Huergo *et al.,* submitted).

Surface plasmon resonance (SPR) measurements. SPR measurements were carried out in an Autolab Esprit instrument (Eco Chemie BV – Netherlands). This SPR technique is equipped with a glass prism (BK7), a planar gold SPR sensor disk (SPR sensor chip), and two measurement channels. For the measurements, a laser diode with a wavelength fixed at 670 nm and a photodiode detector were employed. SPR experiments were performed on a functionalized gold surface obtained by formation of a self-assembled monolayer when allowing 1mM 11-mercaptoundecanoic acid (11-MUA) react in ethanol solution for 24 h. The terminal carboxyl groups of 11-MUA were activated for the formation of NHS-ester groups using 10 mM N-(3 dimethylaminopropyl)-N-ethylcarbodiimidehydrochloride (EDC) and 15 mM N-hydroxy succinimide (NHS) for 15 min. The purified SARS-CoV-2 N-protein (15 $\mu g.ml^{-1}$ in PBS pH 7.4), was covalently immobilized on the sensor chip surface (11–MUA/Au) for 90 min. To prevent non-specific binding, the unbound reactive ester groups were deactivated with 1 mM ethanolamine at pH 8.5 for 5 min. The kinetic of the interaction between the SARS-CoV-2 N protein and antibodies present human serum was evaluated via SPR through the curves of association obtained in the initial reaction times. For these assays, positive and negative human sera for the COVID-19 were combined in a pool (N = 7) and used at indicated dilutions prepared in PBS pH 7.4.

Additional information on ROC curve interpretation. The following information was adapted from the reference [2]. The ROC analysis assesses the performance of the diagnostic method in terms of sensitivity vs specificity for each observed value of the variable assumed as a decision threshold (in the current work, raw optical density cutoff values to differentiate between the control vs disease groups). For each decided cutoff value, different levels of sensitivity and specificity are obtained. By plotting the % sensitivity vs 100 - % specificity at every possible cutoff value the ROC curve is generated (for example see Fig. S4B). A ROC plot for a diagnostic assay with perfect discrimination between negative and positive samples pass through coordinates x = 0 (100 - % specificity) and y = 100 (% sensitivity), in such scenario, the area under the ROC plot would equal to 1. The area under the ROC curve is the most important feature to estimate the accuracy of a diagnostic method. Areas closest to 1 indicate higher diagnostic accuracy.

Additional information of magnetic immunoassay development and performance. The optimized reaction system described in Fig.1 use 2 µl of serum or 5 µl of blood diluted in 0.2 ml of TBST containing 1% of skimmed milk (1/100 dilution). The serum is incubated with N-protein coated magnetic particles for 2 min. The time frame of each step depicted in Fig. 1 was optimized based on empirical observations. Surface plasmon resonance (SPR) confirmed that equilibrium of the antigen-antibody interaction is achieved within 2 min of incubation at 1/100 dilution of COVID-19 positive serum (Fig. S7). It is important to note that adjustment of reagents concentrations and incubation times are critical to obtain data which could be interpreted visually. It worth mentioning that the use of faster kinetic TMB from other vendors produced equivalent OD600nm in less time (typically 3 min). Hence, the total time of the assay can be further reduced to less than 10 min by using faster kinetic TMB solution.

The most appropriated serum dilution to be used on classic and magnetic bead ELISA was determined by serial dilution of a COVID-19 hyperreactive positive serum and a negative control (Fig. S3). The dilutions of 1/1,000 and 1/100 were applied for classic and magnetic bead ELISA, respectively, as these dilutions were about to saturate the signal response of the COVID-19 hyperreactive positive serum while keeping the negative control at reduced cross reaction background level (Fig. S3, dashed lines). The positive serum dilution on the magnetic-bead ELISA showed a response typic of the classic ELISA system [2]. Both classic and magnetic ELISA had similar dynamic ranges with linear response of OD vs reciprocal dilution within 6 data points of the dilution curve (Fig. S3).

The repeatability of the assay was determined in the lab in Brazil in different occasions by running the same positive samples in quintuplicates on the same plate. The CV% was less 3% from low OD values and less than 2% for high OD (n = 5), given the low variation from well to well no duplicates were applied throughout the study. The inter-assay repeatability measured for a high OD sample using different batches of antigen on different days by two different operators showed a CV of 8.5% (n= 25).

The COVID-19 negative cohort used in Brazil where previously evaluated for several parameters including glucose, urea and triglycerides which varied from: 40 to 1052 $mg.dl^{-1}$, 6 to 143 $mg.dl^{-1}$ and 29 to 10,056 $mg.dl^{-1}$, respectively. A total of 9 lipemic sera (triglycerides above 600 $mg.dl^{-1}$) were among these samples. Lipemic serum showed similar response in OD650nm in the magnetic-bead ELISA to other negative samples ($p<0.0001$). Among the positive and negative cohort there were few visible hemolysate samples which showed similar response in OD650nm as those without visible hemolysis ($p<0.0001$). The presence of EDTA, which is commonly used as an anticoagulant in plasm preparation, did not affect the performance of the method in concentrations up to 50 µM (Fig S8A).

The magnetic bead ELISA method was applied for IgM analysis just by changing the secondary antibody (Fig. S8B). The method to investigate IgM operated with an area under the ROC curve of 0.94 whereas 0.98 was obtained for IgG (Fig. S8C to F). The IgM analysis performed with lower accuracy than IgG, 96% specificity and 72% sensitivity where obtained and visual inspection of the data could not be applied for IgM given that the OD for the negative and positive groups overlapped (Fig. S8E). The lower accuracy of IgM analysis in comparison to IgG is well documented in the literature for different diseases [3] and in agreement with our findings.

Supplementary figures Legend

Fig S1. **Details of the purified antigen**. A) SDS-PAGE analysis of purified His-tagged SARS-CoV-2 N protein in lane 1. MW, molecular weight marker (kDa). The gel was Coomassie blue stained. (B) Mascot search peptide mass fingerprint results of the purified protein. The peptides obtained after trypsin digestion were analyzed by MALDI-TOF and searched against virus NCBI database using Mascot PMF search (100 ppm error).

Fig S2. Details of homemade magnetic extractor device. A) Version using a piece of foam as support, the magnified picture shows the details of the nails and magnets used. B) Details of a nail support built on a 3D printer with similar set of magnets as in A and showing how to insert into a 12 wells PCR strip. C) The full system is placed into a lane of the 96 well plate to capture the magnetic beads. D) The beads are fully recovered with the system (see black beads on the tip of the PCR strip tubes) and can be easily transferred from the next lanes. Once the beads are placed in the next lane, the extractor device is lifted leaving the PCR strip on the wells. The PCR strip can be used to mix the beads into the solution. The cost to produce the device is less than US$ 2.

Fig S3. Serum dilution analysis in classic and magnetic bead ELISA. The serum from a COVID-19 positive case and from a negative control were serial diluted and analyzed using classic and magnetic bead ELISA. The data were recorded photographically and expressed as OD values vs reciprocal dilution; the dashed lines indicate the dilution used during this study. Classic ELISA (A and B), magnetic bead ELISA (C and D)

Fig S4. Combined dataset. A) The raw OD650nm for the cohort of samples studied in Brazil and Germany were combined. B) ROC curve of the combined dataset

Fig S5. Point of care validation of the method. A) Examples of photographic register of data obtained at a point of care Unit. C+, positive control serum, C- border line negative control serum used as method cutoff by visual inspection. B, blanks. 1 to 55 samples which were randomized between positive and negatives cases. B) ROC curve of point of care analysis.

Fig S6. The throughout put of the magnetic bead ELISA can be customized on demand. A) A picture of 48 pins magnetic extractor device which can be used to deliver results for 48 samples simultaneously in 12 min. B) Photographic register of data generated in the 48 samples format. C) The results of 17 samples analyzed using the 12 and 48 well format was highly correlated. Data expressed as % of a reference serum.

Fig S7. Surface plasmon resonance analysis of antigen-antibody interaction. The SPR disc was coated with the purified SARS-CoV-2 N protein and allowed to react with the indicated dilutions of a pool of COVID-19 positive sera (A) and pool of negative sera (B). For positive sera at dilution of 1/100, analysis of delta SPR angle vs time suggest that the equilibrium of the antigen-antibody interaction is achieved within 1-2 mins, and longer incubation times were required to reach equilibrium at lower serum dilutions. Effective responses (delta SPR angle) were not obtained for negative sera at dilutions higher than 1/200.

Fig S8. Evaluation of the magnetic bead-based ELISA in the presence of EDTA and to investigate IgM. A) Serum from a COVID-19 positive case was diluted in our reaction system containing the indicated final concentration of EDTA. The color of the reaction did not change despite the presence of EDTA. Thus, EDTA carry over from plasm samples which used EDTA as anticoagulant do not interfere with 6x His tag – Ni^{2+} interaction occurring on the beads. B) The serum from a negative and COVID-19 positive case were analyzed for the presence of IgG and IgM reacting against the SARS-CoV-2 N protein. Blue color indicates positive reaction. The secondary antibody used were anti human IgG or IgM HPR (Thermo Scientific) at 1:3,000 dilution. The OD650nm was expressed as % of a reference for IgG teste (C) and IgM teste (E) for the cohort of samples studied in Brazil. D and F, ROC curve for IgG and IgM teste, respectively

References

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(2) Crowther, J. *The ELISA Guidebook Series Editor*; 2009. https://doi.org/10.1007/978-1-60327-254-4.

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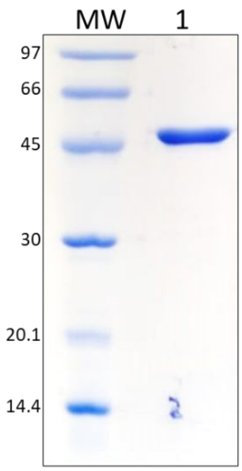
Fig. S1

A



B

Protein View: NCAP_SARS2

Nucleoprotein OS=Severe acute respiratory syndrome coronavirus 2 OX=2697049

Database:	SARS-CoV-2
Score:	137
Expect:	8.2e-13
Monoisotopic mass (M_r):	45598
Calculated pI:	10.07

Sequence similarity is available as an NCBI BLAST search of NCAP_SARS2 against nr.

Search parameters

Enzyme:	Trypsin: cuts C-term side of KR unless next residue is P.
Variable modifications:	Oxidation (M)
Mass values searched:	22
Mass values matched:	13

Protein sequence coverage: 40%

Matched peptides shown in ***bold red***.

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  1 MSDNGPQNQR NAPRITFGGP SDSTGSNQNG ERSGARSKQR RPQGLPNNTA
 51 SWFTALTQHG KEDLKFPRGQ GVPINTNSSP DDQIGYYRRA TRRIRGGDGK
101 MKDLSPRWYF YYLGTGPEAG LPYGANKDGI IWVATEGALN TPKDHIGTRN
151 PANNAAIVLQ LPQGTTLPKG FYAEGSRGGS QASSRSSSRS RNSSRNSTPG
201 SSRGTSPARM AGNGGDAALA LLLLDRLNQL ESKMSGKGQQ QQGQTVTKKS
251 AAEASKKPRQ KRTATKAYNV TQAFGRRGPE QTQGNFGDQE LIRQGTDYKH
301 WPQIAQFAPS ASAFFGMSRI GMEVTPSGTW LTYTGAIKLD DKDPNFKDQV
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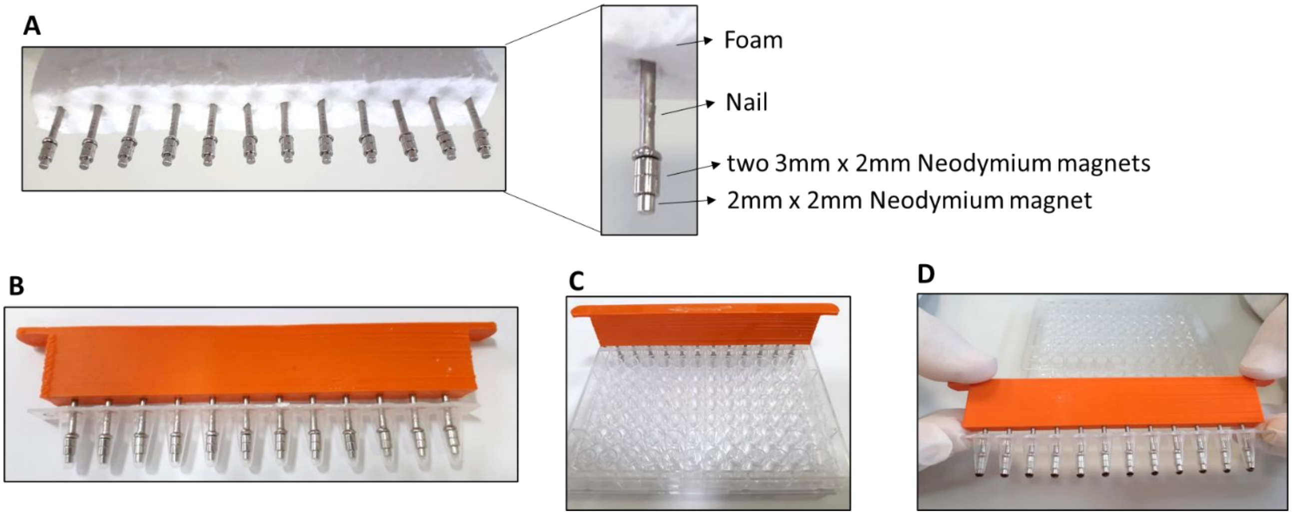
Fig. S2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Nail
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2mm x 2mm Neodymium magnet
B
C
D

Fig. S3

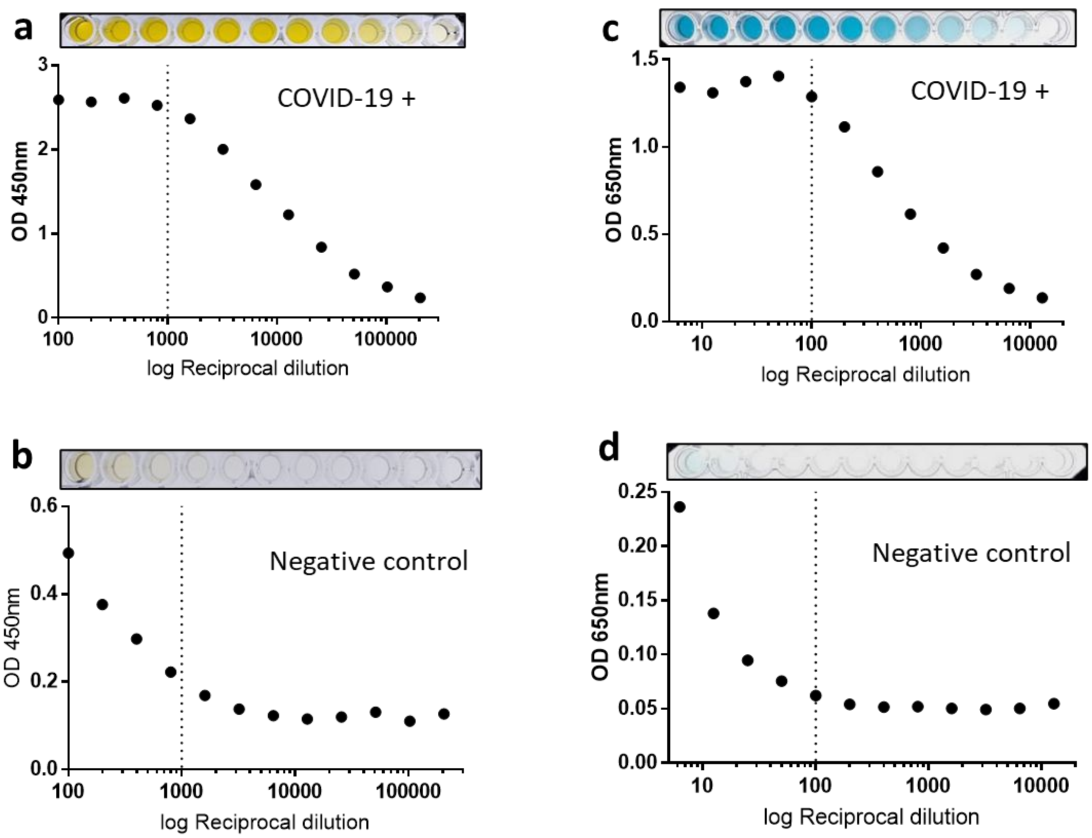

a
OD 450nm
3
2
1
0
100
1000
10000
100000
log Reciprocal dilution
COVID-19 +
c
OD 650nm
1.5
1.0
0.5
0.0
10
100
1000
10000
log Reciprocal dilution
COVID-19 +
b
OD 450nm
0.6
0.4
0.2
0.0
100
1000
10000
100000
log Reciprocal dilution
Negative control
d
OD 650nm
0.25
0.20
0.15
0.10
0.05
0.00
10
100
1000
10000
log Reciprocal dilution
Negative control

Fig. S4

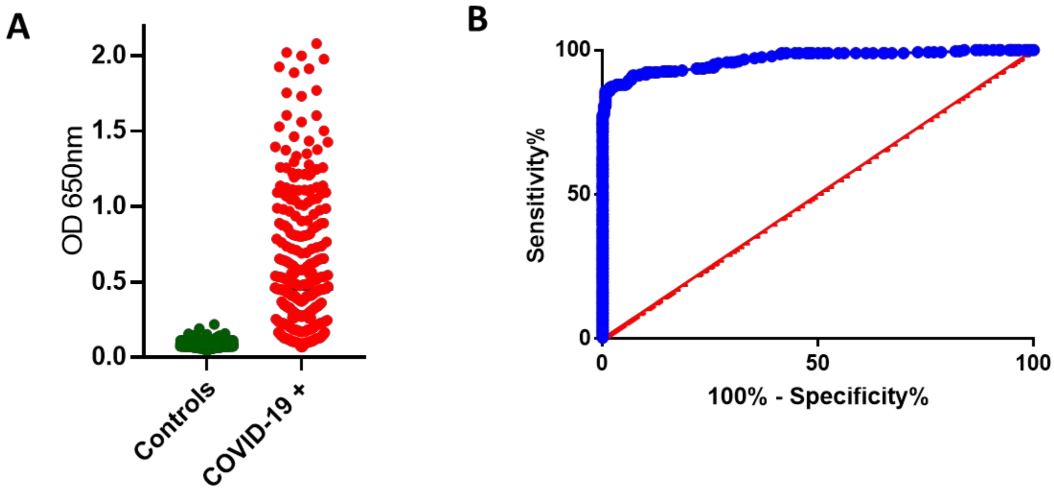

A
2.0
1.5
1.0
0.5
0.0
OD 650nm
Controls
COVID-19 +
B
100
50
0
Sensitivity%
0
50
100
100% - Specificity%

Fig. S5

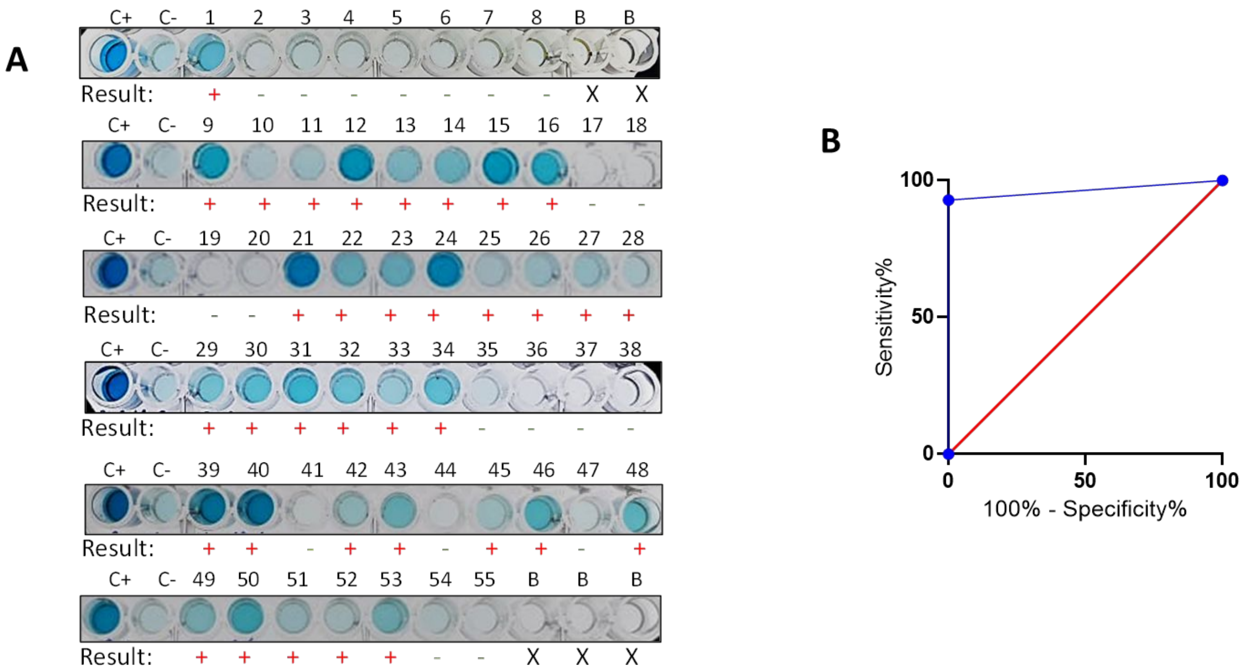
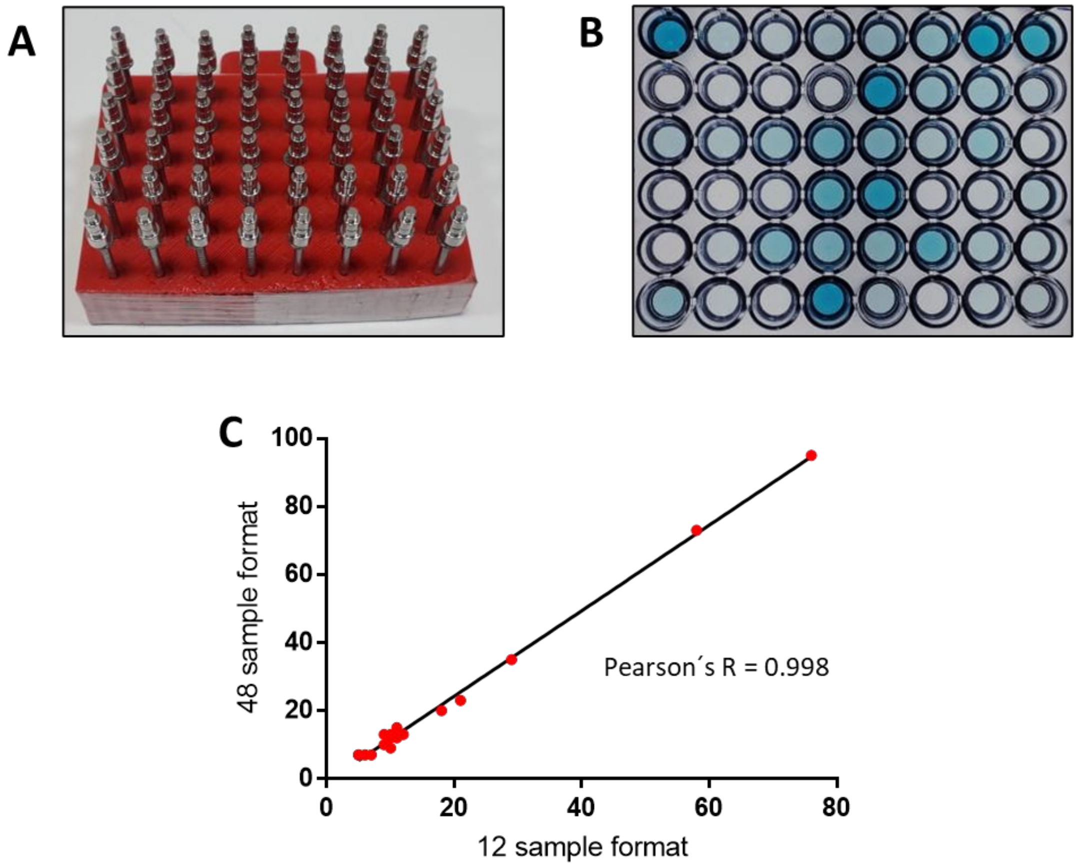

A
C+ C- 1 2 3 4 5 6 7 8 B B
Result: + - - - - - - - X X
C+ C- 9 10 11 12 13 14 15 16 17 18
Result: + + + + + + + + - -
C+ C- 19 20 21 22 23 24 25 26 27 28
Result: - - + + + + + + + +
C+ C- 29 30 31 32 33 34 35 36 37 38
Result: + + + + + + - - - -
C+ C- 39 40 41 42 43 44 45 46 47 48
Result: + + - + + - + + - +
C+ C- 49 50 51 52 53 54 55 B B B
Result: + + + + + - - X X X
B
Sensitivity%
100
50
0
100% - Specificity%

Fig. S6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B
C
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60
40
20
0
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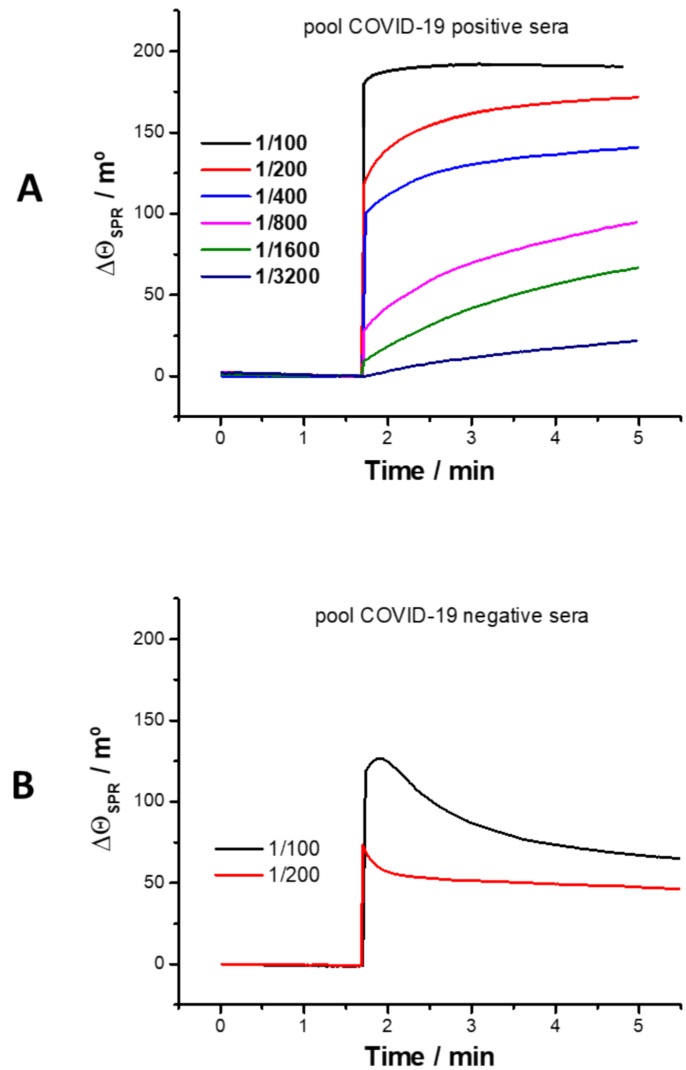
0
20
40
60
80
12 sample format

Fig. S7


A
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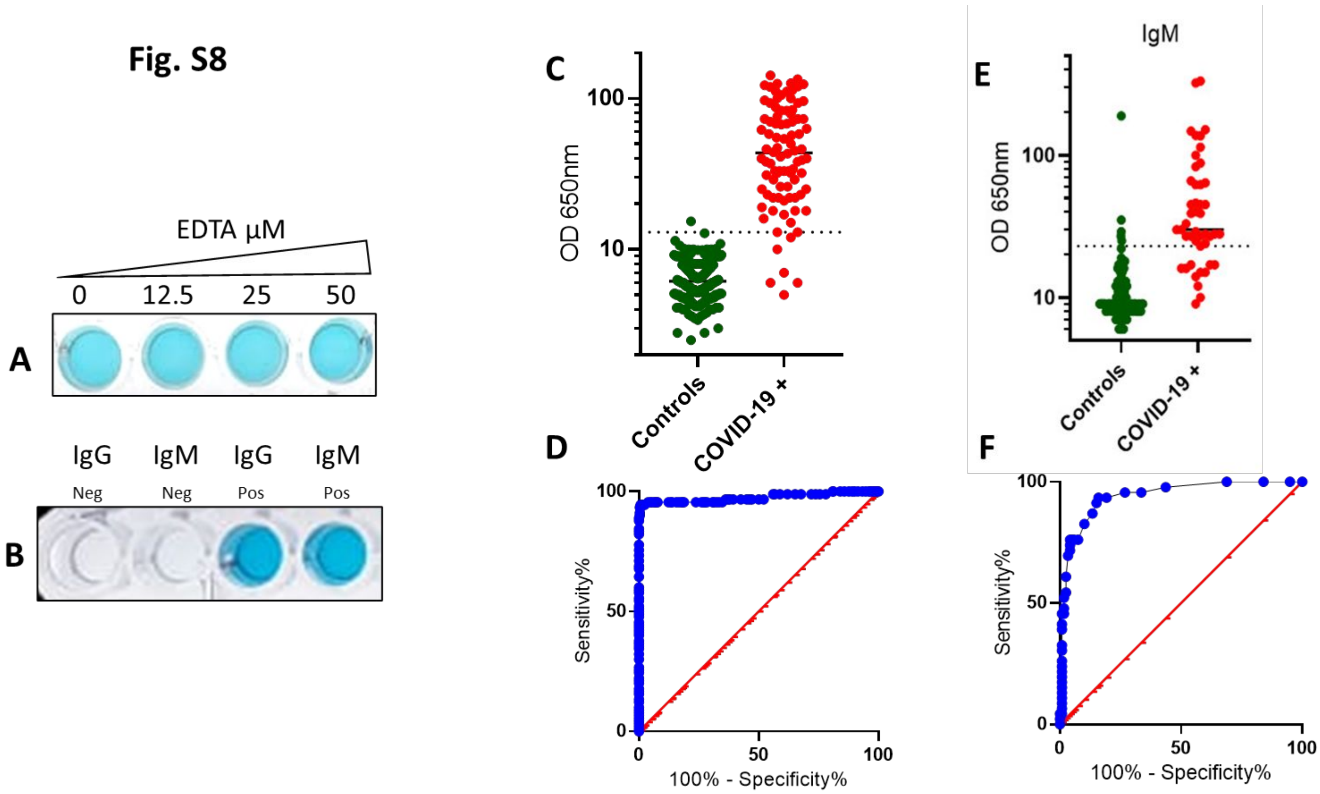
$\Delta\Theta_{SPR}$ / m°
1/100
1/200
1/400
1/800
1/1600
1/3200
Time / min
B
pool COVID-19 negative sera
$\Delta\Theta_{SPR}$ / m°
1/100
1/200
Time / min

Fig. S8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25
50
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IgG
Neg
IgM
Neg
IgG
Pos
IgM
Pos
B
C
OD 650nm
100
10
Controls
COVID-19 +
D
Sensitivity%
100% - Specificity%
IgM
E
F